

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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Full proposal: research.ekdynamics.science/proposal.pdf

The problem. The reachable set of an atmospheric entry vehicle, that is, every state it can be driven to under admissible control, is what a safety analysis needs bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in the state dimension and capped in practice near four or five states [1]. A rigid-body entry vehicle carries thirteen states, and more once the heat load, ablation and aerothermoelastic deformation are coupled, so the grid method stops before the real problem starts. A polynomial barrier whose coefficients come from an interior-point solver has analytical *form*, but it inherits the solver’s accuracy and is not by itself a certificate. There are therefore two problems to solve, not one: the dimension of the state, and the provenance of the guarantee. The vehicle and its corridor enter only as data, so the bound sought is for hypersonic entry in general.

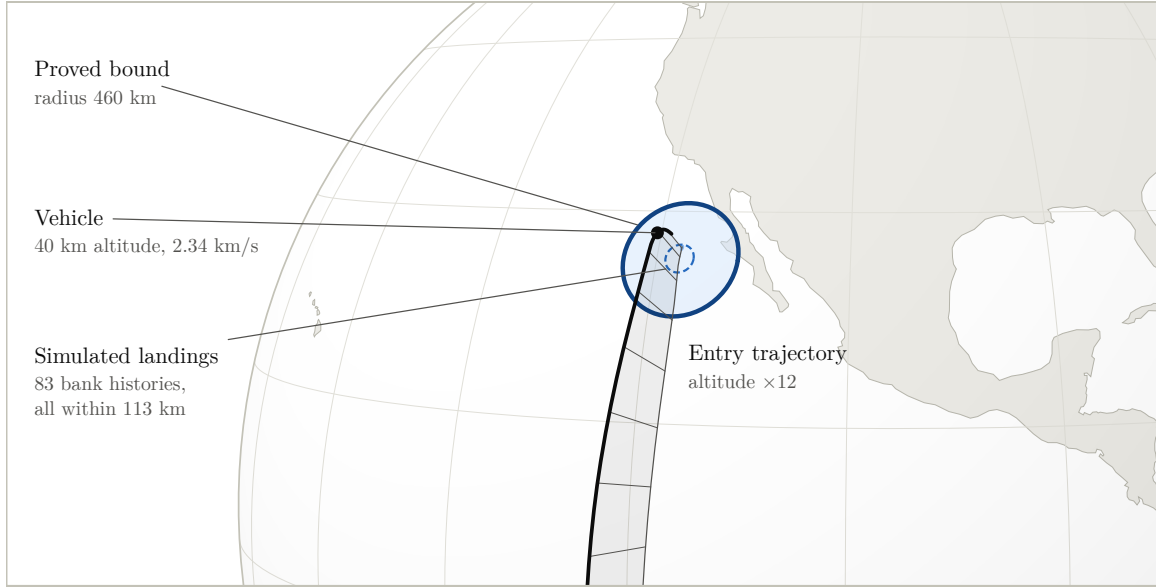


Figure 1: A first bound, on an example: a lunar-return skip entry simulated from a published entry state [2], with altitude drawn twelve times too large. From the state marked, every bank history that stays below 40 km lands inside the solid circle; this is proved (see *A first bound*). Simulated bank histories land inside the dashed one. The program is to close the gap.

The approach. We embed the flight dynamics in a space of occupation measures. The nonlinear flight mechanics $\dot{x} = f(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$, and an outer bound on the reachable set is certified by a dual pair of functions. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [3]. Because the aerodynamics are transcendental and known only to within tolerances, we enclose them in intervals. The dynamics become a differential inclusion, evaluated in validated interval arithmetic, with no numerical optimization as the source of the guarantee.

The new content is what happens in the thin-atmosphere limit $\varepsilon = H_s/R_e \rightarrow 0$, where H_s is the scale height of the atmosphere and R_e the radius of the planet. In this limit a trajectory separates into three motions: arcs above the atmosphere, governed by gravity alone; the equilibrium glide, on which lift balances weight less centrifugal relief; and atmospheric *passes*, in which the vehicle dips into dense air, turns, and sheds a fixed share of its energy in a vanishing time. On the flight corridor the occupation measures remain bounded and converge along subsequences, but their limit forgets the passes: a measure weighs each interval of time by its length, and in the limit a pass has none. Since a reachable set is a support and not an integral, what has no mass still counts.

We therefore read the loss where it occurs, on the energy dissipation. Along every trajectory, for every control and every admissible model of the aerodynamics, the mechanical energy in the rotating frame decreases at the rate of the drag power, $\dot{\mathcal{E}} = -P \leq 0$, so the total dissipation is bounded uniformly in ε . We rescale time by $\sqrt{\varepsilon}$, as in the classical variables of Chapman [4]. Since the rescaled time axis is unbounded, compactness is lost, and the concentration-compactness principle of Lions [5] classifies the normalized dissipation densities into vanishing, compactness, or dichotomy. We expect these to be the glide, a single pass, and passes that separate.

Rather than restore compactness, we extract the profile decomposition that iterating the dichotomy provides, as in the theory of nonlinear dispersive equations [6]:

- a low-dimensional *skeleton*, which is what survives the limit: the arcs above the atmosphere and the equilibrium glide;
- one *boundary-layer bubble* per atmospheric pass, the analogue of a concentrating profile: the pass rescaled to its own width, whose net effect on the slow coordinates is a jump;
- a *residual*, which is what the finitely many profiles do not capture, to be bounded explicitly in the corridor metric.

The number of bubbles is bounded *a priori* by the energy budget $\Delta\mathcal{E}$: a pass that dissipates at least δ can occur at most $K_\delta \leq \Delta\mathcal{E}/\delta$ times. Skeleton and bubbles together form a limit system $\mathcal{H}_\delta$ in a few slow coordinates, which flows along the arcs and the glide and jumps at each pass. The target statement is

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\Pi(\mathcal{R}_{\mathcal{H}_\delta}(T))}_{\text{skeleton and bubbles}} \oplus \underbrace{\mathbb{B}(\varrho(\varepsilon, \delta))}_{\text{residual}},$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. Here $\mathcal{R}_\varepsilon(T)$ is the reachable set of the full system, Π reads a state of the limit system back in the full coordinates, and $\mathbb{B}(\varrho)$ is a ball in the corridor metric. The count K_δ and the radius ϱ are to be computable in closed form from the corridor bounds and the aerodynamic interval widths alone. Compactness provides this structure but no constants; those are to come from direct estimates: the energy identity, a matching estimate at each pass, and averaging over the oscillations that do not concentrate.

A first bound. The first of these estimates can be carried out at once. Suppose that the vehicle stays below a ceiling where the density is at least ρ_c , that the air thickens below it at least as fast as an exponential atmosphere of scale height H under gravity g , and that the ballistic coefficient is at most β . The ground track advances no faster than the vehicle, so integrating the energy identity while the speed falls from V_0 to V_f bounds its length s :

$$s \leq \frac{2\beta}{\rho_c} \left[\ln \frac{V_0}{V_f} + \frac{gH}{V_f^2} \right]$$

for every control and every admissible model of the aerodynamics; the proof is in the full proposal. In the example of Figure 1 this is 460 km, where simulated bank histories reach 113 km. The bound is elementary and far from sharp, and the way it fails is the point: it is proportional to $1/\rho_c$, so it says nothing about a trajectory that leaves the atmosphere between two passes. This is why each piece must be estimated on its own scale.

Open problems. The central theorem is open; only the bound above is claimed. The following problems remain.

1. **The limit.** To show that the limit of the occupation measures satisfies the Liouville equation of the limit system, with a jump term carried by the passes, uniformly over the admissible family.
2. **Composition of passes.** To show that a trajectory with several passes factors through the individual passes and the flows between them, with an explicit matching remainder. The classical matched expansions give its form for one trajectory [7]; the statement needed is uniform over the family.
3. **Separation of scales.** The attitude and elastic motions do not concentrate. They are persistent oscillations, as is the oscillation about the glide, and in a profile decomposition such motions belong to the remainder [6]. The tool for them is averaging [8], but their frequencies cross during every pass, and the available results give orders and not bounds, and hold through a resonance only outside a set of small measure, which an outer bound cannot discard. What is needed is an averaging theorem with explicit constants, uniform over the admissible family and valid through resonance.

The third problem is the critical path, and it is the part of the program for which I am looking for an advisor.

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